

Hari's Practical AI Engineer Roadmap

Goal: Transition from Automation Engineer to AI Engineer using a practical, project-based roadmap focused on modern LLM engineering and AWS transition later.

Phase 1 — Python + AI Foundations

Learn:

- Python basics
- Functions
- File handling
- Exceptions
- JSON
- APIs

Build:

- API caller
 - JSON parser
 - Mini chatbot
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Phase 2 — LLM Basics

Learn:

- Prompts
- Tokens
- Temperature
- Context windows
- Hallucinations

Build:

- Text summarizer
 - Prompt playground
 - Q&A; app
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Phase 3 — RAG Foundations

Learn:

- PDF parsing
- Chunking
- Embeddings
- Semantic search
- Vector DB

Build:

- PDF chatbot
- Semantic search engine
- Multi-document retrieval

Phase 4 — Backend AI Engineering

Learn:

- FastAPI
- REST APIs
- Async basics
- Streaming

Build:

- AI API backend
 - Chatbot backend service
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Phase 5 — Frontend AI Apps

Learn:

- Simple AI UI building
- Frontend-backend connection

Build:

- Streamlit AI app
 - PDF upload chatbot
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Phase 6 — Advanced AI Engineering

Learn:

- Agents
- Memory
- Evaluation
- Observability

Build:

- AI agents
 - Autonomous workflows
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Phase 7 — AWS AI Engineering

Learn:

- AWS basics
- S3
- Lambda
- Bedrock
- Docker

- Build:**
- Cloud RAG systems
 - Enterprise AI deployment
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Recommended Beginner Tool Stack

Category	Tool
LLM	Groq
Parsing	PyMuPDF
Orchestration	LangChain
Embeddings	Sentence Transformers
Vector DB	FAISS
Backend	FastAPI
Frontend	Streamlit
IDE	VS Code

Core Principle:
Learn → Build → Break → Debug → Improve.

Avoid endless tutorial watching. Focus on consistently building practical AI systems.